
Amazon ECS – Creating an ECS Cluster

What is an ECS Cluster?

An ECS cluster is a logical grouping of ECS resources (EC2 or Fargate tasks).

Steps to Create an ECS Cluster (Console)

Using Fargate (recommended for most workloads):

1. Open the **ECS Console**.
2. Click "**Clusters**" > "**Create Cluster**".
3. Choose "**Networking only (Fargate)**".
4. Enter **Cluster Name**.
5. Optional: Enable **Container Insights**.
6. Click **Create**.

Using EC2 Launch Type:

1. Select **EC2 Linux + Networking**.
2. Choose:
 - **Instance type**
 - **Key pair**
 - **VPC/Subnet**
3. ECS will auto-provision EC2 instances with ECS Agent.

Considerations:

Decision Point	Recommendation
Launch Type	Fargate (unless you need custom EC2)
Networking	Use VPC with private subnets
Insights	Enable Container Insights
Cluster Name	Follow naming convention (env-app)

Amazon ECS – Creating Task Definitions

What is a Task Definition?

A JSON blueprint for running containers. It defines:

- Image
 - CPU & Memory
 - Ports
 - IAM Role
 - Environment Variables
 - Logging
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Steps to Create a Task Definition (Console)

1. Go to **Task Definitions** > **Create new Task Definition**.
2. Choose **Fargate** or **EC2**.

3. Enter:

- **Task Name**
- **Task Role** (IAM Role for the app)
- **Execution Role** (IAM Role for ECS to pull image, send logs)

4. Add Container:

- **Container name**
- **Image URL** (e.g. `123456789012.dkr.ecr.region.amazonaws.com/my-app:latest`)
- **Port mappings** (e.g. 80)
- **CPU / Memory**
- **Log Configuration:** awslogs or FireLens

5. Optional:

- Environment variables
- Secrets from **Secrets Manager**

6. Click **Create**

Best Practices

Area	Practice
Image Source	Use ECR with versioned tags
Logging	Enable <code>awslogs</code> driver for all containers
Secrets	Use Secrets Manager or SSM with secure references
Resource Limits	Set precise CPU and memory to optimize cost

Network Mode Use **aws**vpc for task-level ENI and isolation

Amazon ECS – Creating Services and Tasks



What is a Service?

An ECS Service ensures that a specified number of task instances are always running and optionally integrates with load balancers.



Steps to Create a Service

1. Go to **Clusters** > Your Cluster > **Services** > **Create**.
2. Select:
 - **Launch type**: Fargate or EC2
 - **Task definition** and **revision**
 - **Service name**
 - **Number of tasks** to run
3. Networking:
 - **VPC + Subnets** (use private subnets with NAT)
 - **Security Group** (allow HTTP/HTTPS/Custom ports)
 - **Assign Public IP**: Only if using public subnets
4. Load Balancer (optional):
 - Use **Application Load Balancer**
 - Select **Target Group**
5. Auto Scaling (optional):

- Enable service auto scaling based on CPU/memory

6. Click **Create Service**

Running One-Time Tasks

To run a **standalone ECS Task** (e.g. cron job or batch process):

1. Go to **Clusters** > Select Cluster > **Tasks** > **Run New Task**
 2. Choose:
 - Launch type (Fargate)
 - Task definition and revision
 - Subnet and Security Group
 3. Click **Run Task**
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Amazon ECS – Task Placement Strategies

What is Task Placement?

For **EC2 launch type**, ECS decides **which EC2 instance** to place a task on.

Placement Strategies

Strategy	Description
binpack	Packs tasks onto least number of instances (cost-efficient).
spread	Spreads tasks across multiple instances/AZs (high availability).
random	Places randomly among instances.

You can combine strategies. For example:

```
[
  {
    "type": "spread",
    "field": "attribute:ecs.availability-zone"
  },
  {
    "type": "binpack",
    "field": "memory"
  }
]
```

Placement Constraints

Constraint Type

Example Use Case

distinctInstance Run each task on a different EC2 instance

memberOf Custom instance selection via expressions (e.g., AZs, instance types)

```
[
  {
    "type": "memberOf",
    "expression": "attribute:ecs.instance-type == t3.medium"
  }
]
```

Placement for Fargate

- No placement strategy needed.
- AWS handles compute isolation and placement automatically.

Summary Checklist

ECS Component

Key Decisions / Actions

Cluster	Choose Fargate for simplicity and scalability
Task Definition	Define image, roles, logs, and secure environment variables
Service	Use ALB for traffic, private subnets, and auto scaling
Task Placement	Use binpack/spread strategies on EC2, not needed for Fargate
